EECE7352 Computer Architecture

Homework 1

Sakthivel P. Sivakumar

NUID: 002312294

Part B:

Linpack Benchmark

The LINPACK benchmark is designed to measure the **floating-point performance** of a system by solving a dense system of linear equations using LU factorization. It is particularly sensitive to compiler optimizations, math libraries, and the efficiency of floating-point extensions such as SSE/AVX on x86-64.

lota.coe.neu.edu_gcc_O0_out.txt

PartB > runs > lota.coe.neu.edu_gcc_O0_out.txt

```
22 September 2025 08:27:47 PM

LINPACK_BENCH
  C version

  The LINPACK benchmark.
  Language: C
  Datatype: Double precision real
  Matrix order N               = 1000
  Leading matrix dimension LDA = 1001

     Norm. Resid      Resid           MACHEP         X[1]          X[N]

        6.491510       0.000000    2.220446e-16       1.000000        1.000000

      Factor     Solve      Total     MFLOPS       Unit      Cray-Ratio

   1.205946   0.003697   1.209643  552.780173   0.003618  21.600768

LINPACK_BENCH
  Normal end of execution.

22 September 2025 08:27:48 PM
```

lota.coe.neu.edu_gcc_O1_out.txt

PartB > runs > lota.coe.neu.edu_gcc_O1_out.txt

```
22 September 2025 08:28:15 PM

LINPACK_BENCH
  C version

  The LINPACK benchmark.
  Language: C
  Datatype: Double precision real
  Matrix order N               = 1000
  Leading matrix dimension LDA = 1001

     Norm. Resid      Resid           MACHEP         X[1]          X[N]

        6.491510       0.000000    2.220446e-16       1.000000        1.000000

      Factor     Solve      Total     MFLOPS       Unit      Cray-Ratio

   0.308207   0.000846   0.309053  2163.598692   0.000924   5.518804

LINPACK_BENCH
  Normal end of execution.

22 September 2025 08:28:16 PM
```

lota.coe.neu.edu_gcc_O2_out.txt

PartB > runs > lota.coe.neu.edu_gcc_O2_out.txt

```
22 September 2025 08:29:21 PM

LINPACK_BENCH
  C version

  The LINPACK benchmark.
  Language: C
  Datatype: Double precision real
  Matrix order N               = 1000
  Leading matrix dimension LDA = 1001

     Norm. Resid      Resid           MACHEP         X[1]          X[N]

        6.491510       0.000000    2.220446e-16       1.000000        1.000000

      Factor     Solve      Total     MFLOPS       Unit      Cray-Ratio

   0.306617   0.000840   0.307457  2174.829868   0.000920   5.490304

LINPACK_BENCH
  Normal end of execution.

22 September 2025 08:29:21 PM
```

lota.coe.neu.edu_gcc_O3_out.txt

PartB > runs > lota.coe.neu.edu_gcc_O3_out.txt

```
22 September 2025 08:29:47 PM

LINPACK_BENCH
  C version

  The LINPACK benchmark.
  Language: C
  Datatype: Double precision real
  Matrix order N               = 1000
  Leading matrix dimension LDA = 1001

     Norm. Resid      Resid           MACHEP         X[1]          X[N]

        6.491510       0.000000    2.220446e-16       1.000000        1.000000

      Factor     Solve      Total     MFLOPS       Unit      Cray-Ratio

   0.232429   0.000619   0.233048  2869.222935   0.000697   4.161571

LINPACK_BENCH
  Normal end of execution.

22 September 2025 08:29:47 PM
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

ssh - HW1

```
sakthivelps@degrees:~/EECE7352/sakthivel/HW1/PartA/ARM_V8_GCC/profiles$
```

Table:

Optimization	Factor (s)	Solve (s)	Total (s)	MFLOPS	Cray-Ratio
O0	1.205946	0.003697	1.209643	552.780173	21.600768
O1	0.308207	0.000846	0.309053	2163.598692	5.518804
O2	0.306617	0.00084	0.307457	2174.829868	5.490304
O3	0.232429	0.000619	0.233048	2869.222935	4.161571

Metrics reported in the Benchmark

The benchmark reports several **metrics**:

- **X[1], X[N]**: spot checks of the solution vector for sanity validation.
- **Factor, Solve, Total times**: timing components. *Factor* is the LU decomposition stage, *Solve* is the forward/back substitution, and *Total* is the combined execution time.
- **MFLOPS (Millions of Floating-Point Operations per Second)**: the **primary performance metric**. It reflects how quickly the CPU executed the floating-point operations for the chosen matrix size.
- **Cray Ratio**: a historical metric comparing performance against a Cray baseline system.

From the results collected across different -O levels, the impact of compiler optimizations is clear. At **-O0**, performance is very low **(~ 553 MFLOPS**). Moving to **-O1** and **-O2**, the MFLOPS rises sharply into the thousands (~2,100 MFLOPS at -O2), showing the benefits of instruction scheduling and partial vectorization. At **-O3**, the compiler fully leverages loop unrolling, inlining, and SIMD instructions, pushing throughput to nearly **2,900 MFLOPS**. This is a more than **5X improvement** compared to the unoptimized build.

In summary, LINPACK is not just a measure of raw clock speed but of the system's ability to sustain **floating-point intensive workloads** using optimized code paths and vectorized math instructions. The benchmark demonstrates how crucial compiler optimizations are for unlocking the real floating-point potential of modern x86-64 hardware.